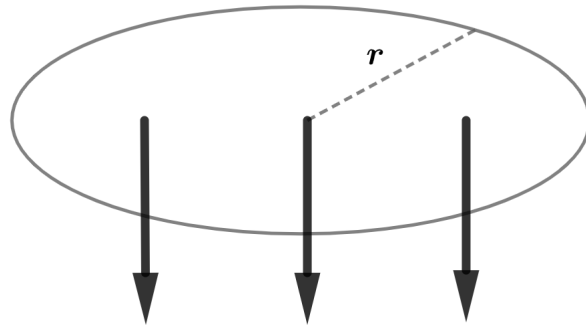


2020A F=ma Exam: Problem 11

Kevin S. Huang



The tensile strength is given by

$$P = \frac{T_{\max}}{A} = \frac{mg}{\pi r^2}$$

so

$$m = \frac{\pi r^2}{g} P = \frac{\pi (1 \text{ cm})^2}{(10 \text{ N/m})} (415 \text{ MPa}) = 13,000 \text{ kg}$$

and the answer is .